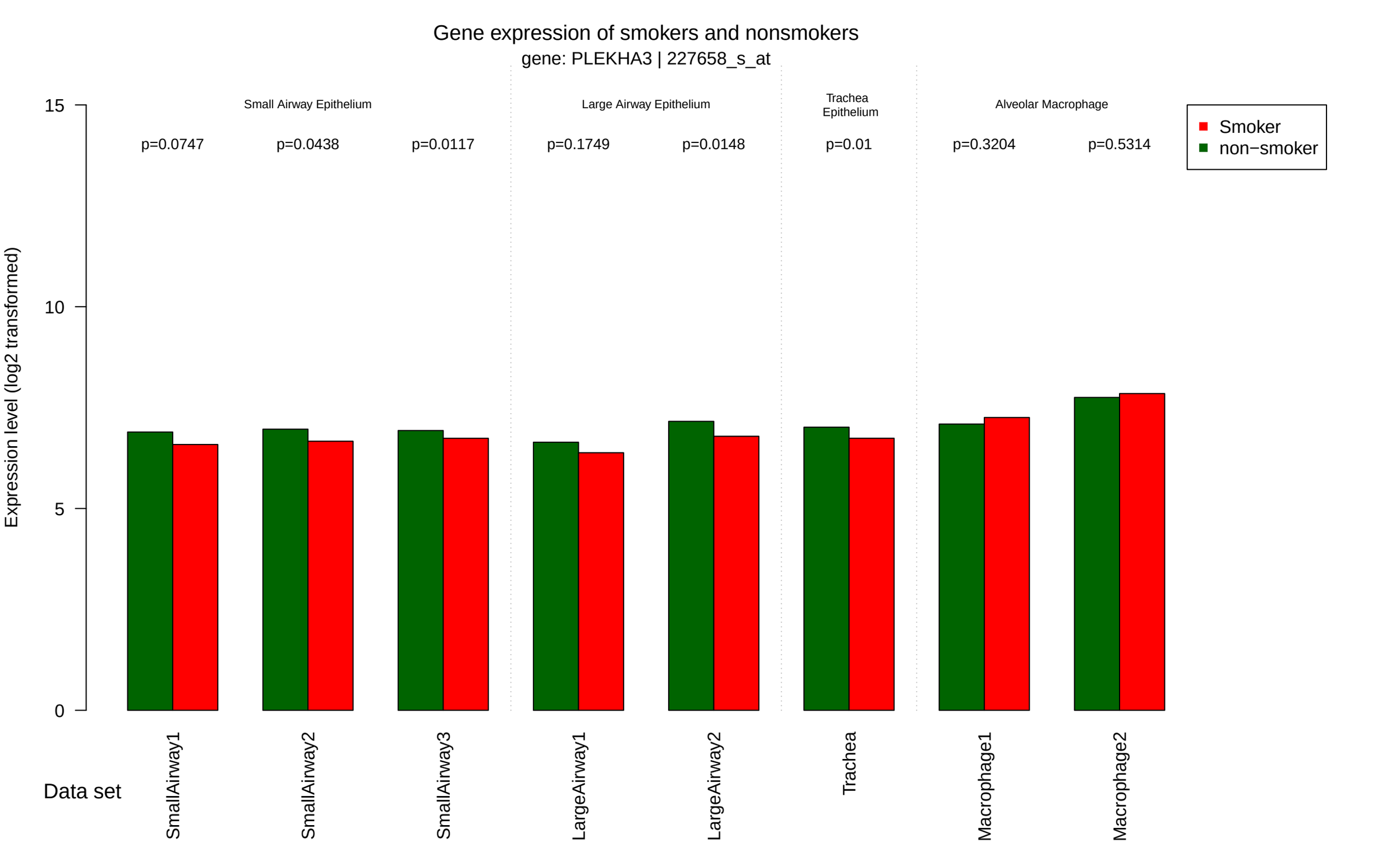
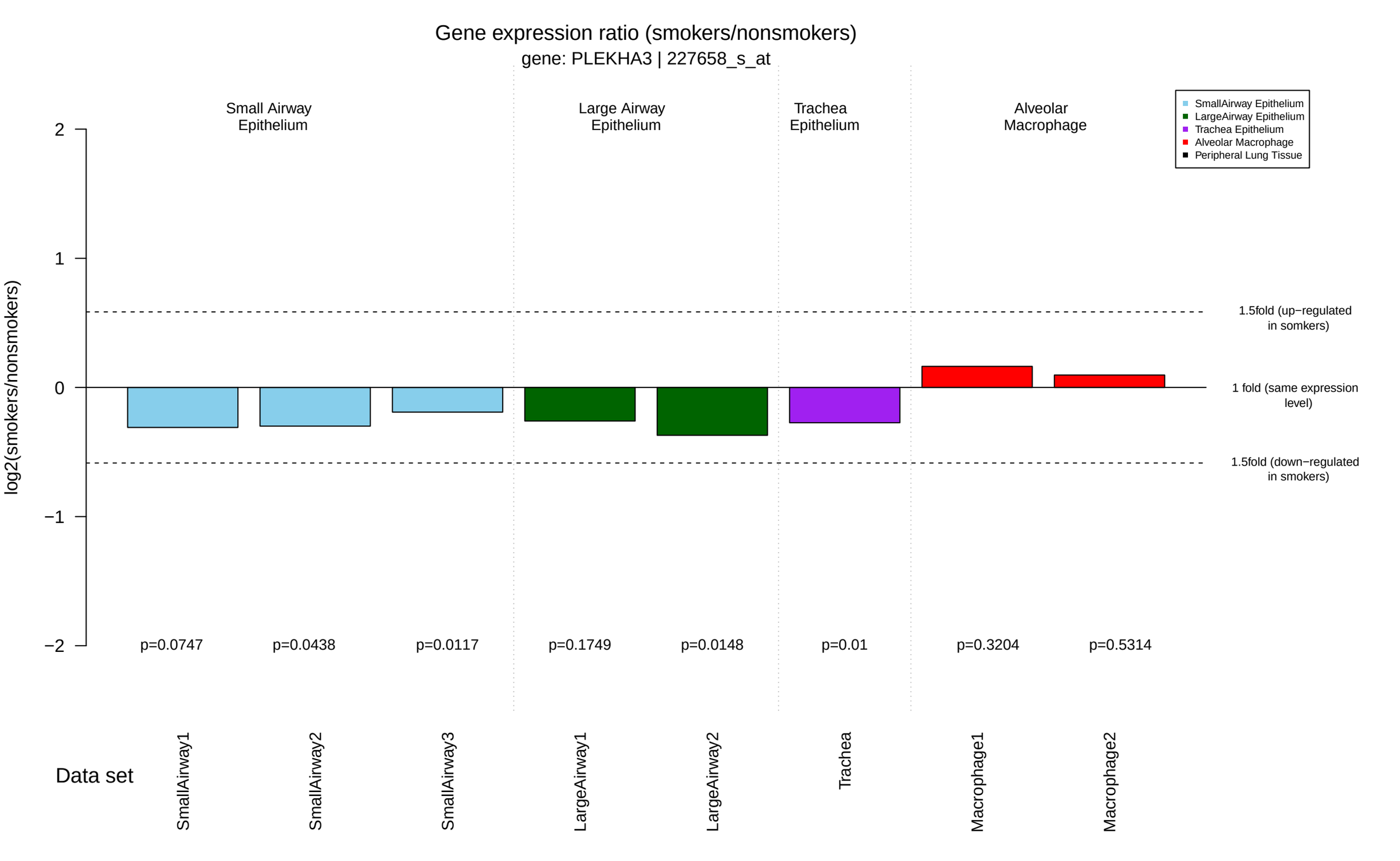
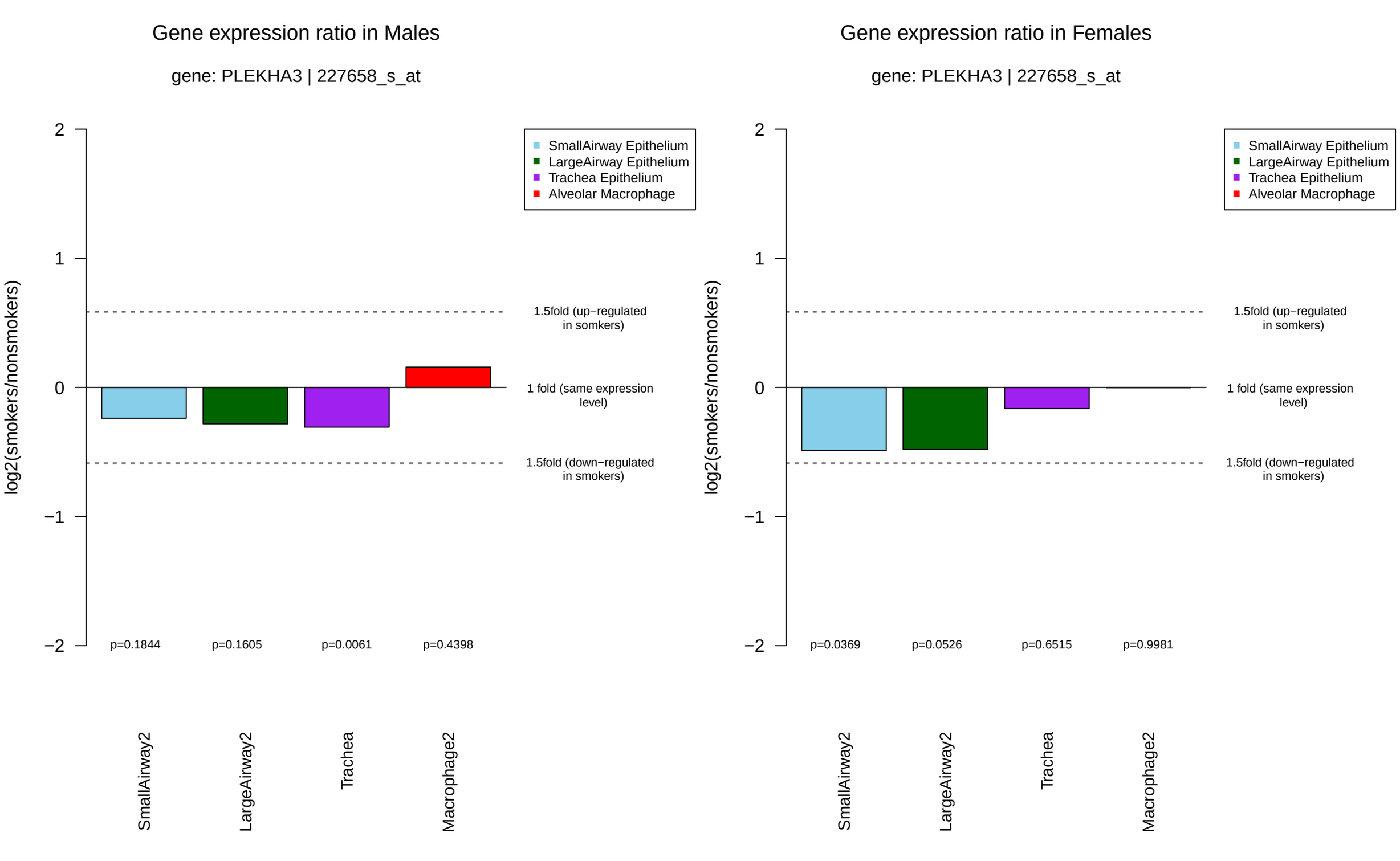
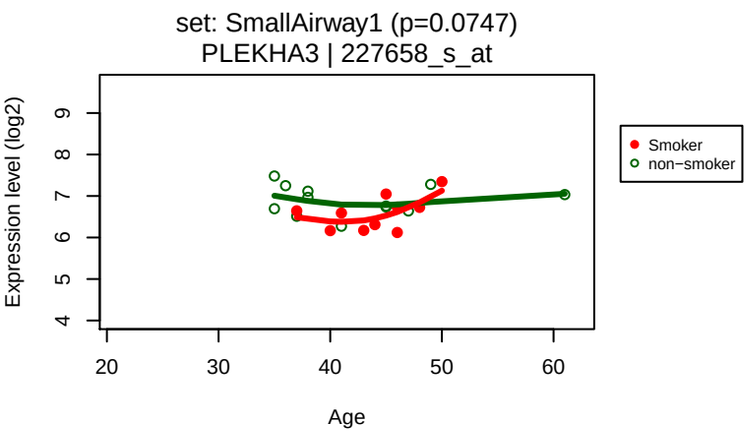
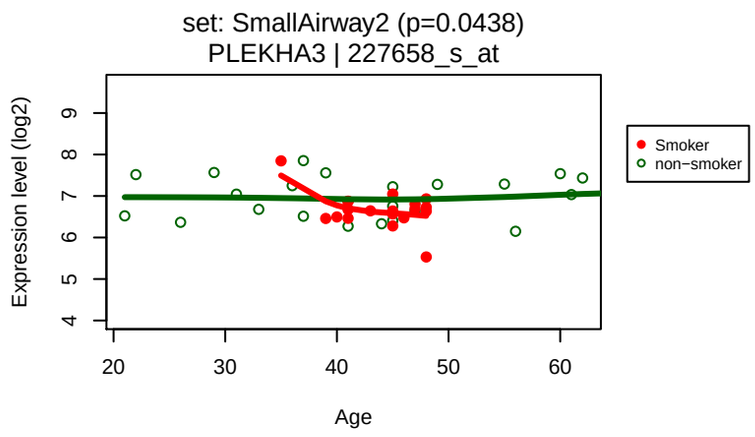
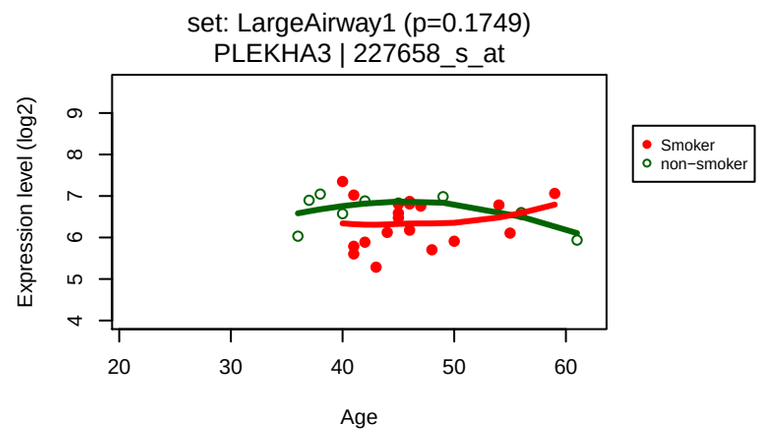

Gene expression of smokers and nonsmokers
gene: PLEKHA3 | 227658_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0747
p=0.0438
p=0.0117
p=0.1749
p=0.0148
p=0.01
p=0.3204
p=0.5314
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

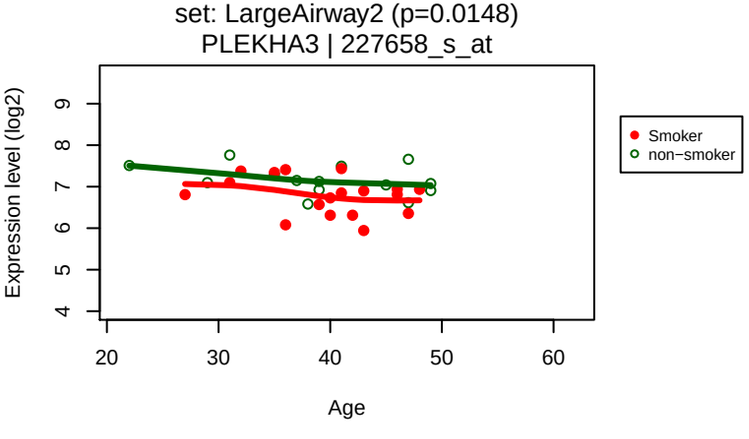

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHA3 | 227658_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0747
p=0.0438
p=0.0117
p=0.1749
p=0.0148
p=0.01
p=0.3204
p=0.5314
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

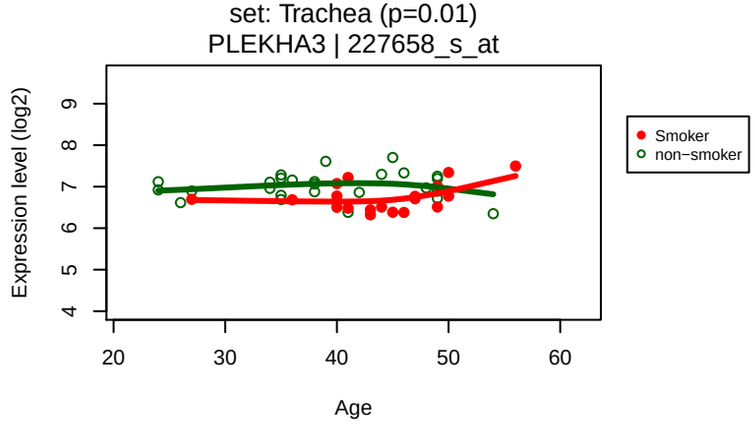

Gene expression ratio in Males
gene: PLEKHA3 | 227658_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1844
p=0.1605
p=0.0061
p=0.4398
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHA3 | 227658_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0369
p=0.0526
p=0.6515
p=0.9981
SmallAirway2
LargeAirway2
Trachea
Macrophage2

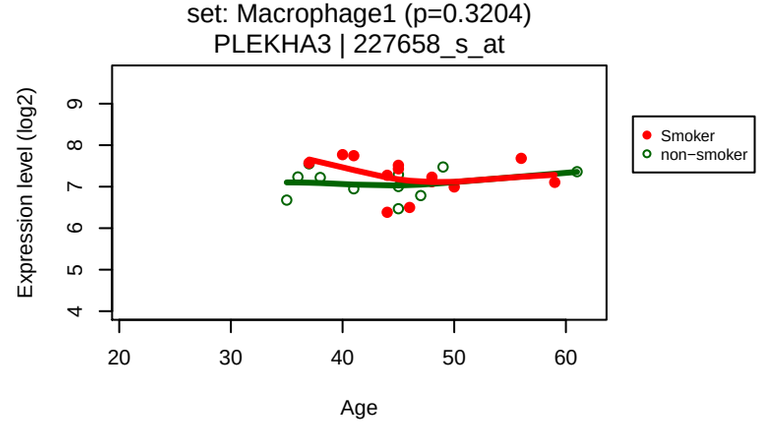

set: SmallAirway1 (p=0.0747)
PLEKHA3 | 227658_s_at
Expression level (log2)
Age
Smoker
non−smoker

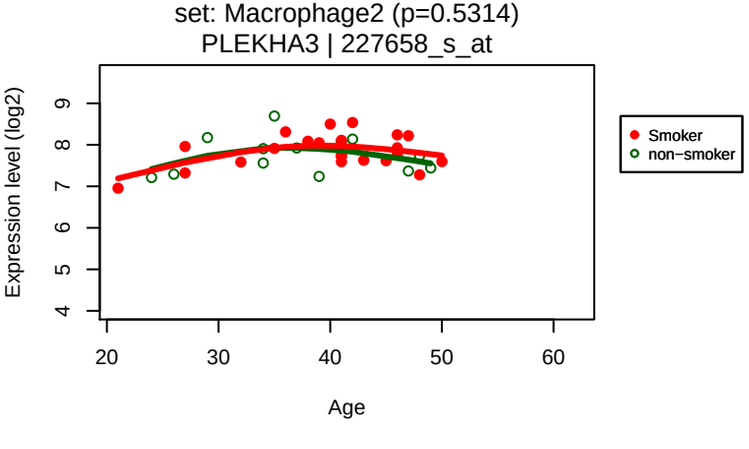

set: SmallAirway2 (p=0.0438)
PLEKHA3 | 227658_s_at
Expression level (log2)
Age
Smoker
non−smoker

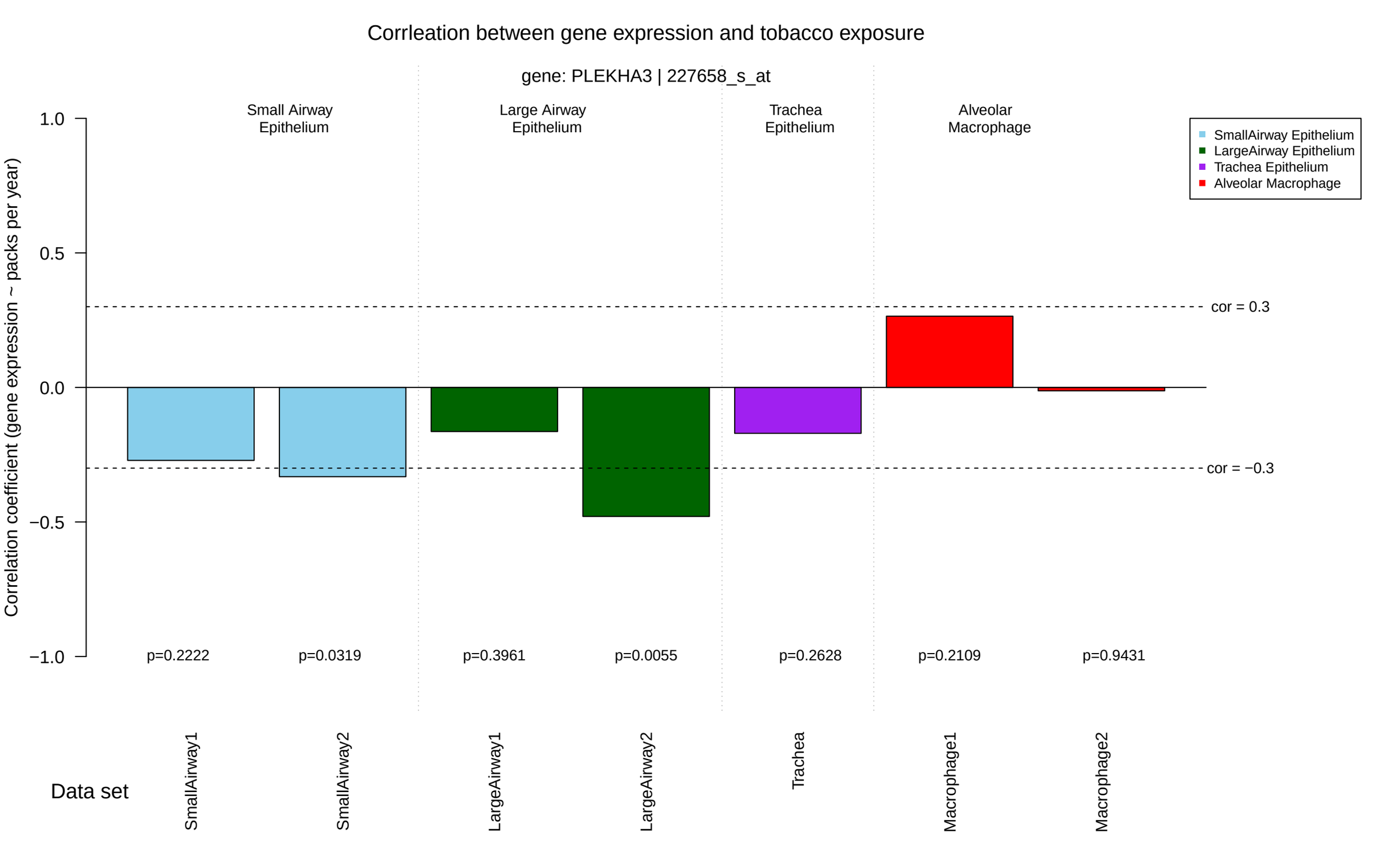

set: LargeAirway1 (p=0.1749)
PLEKHA3 | 227658_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0148)
PLEKHA3 | 227658_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.01)
PLEKHA3 | 227658_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3204)
PLEKHA3 | 227658_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5314)
PLEKHA3 | 227658_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHA3 | 227658_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2222
p=0.0319
p=0.3961
p=0.0055
p=0.2628
p=0.2109
p=0.9431
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2